Curriculum Vitae

Daria Bogatova

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Scientific Focus

I am a Ph.D. candidate in Neurobiology at Boston University **working at the intersection of optical engineering, systems neuroscience, and computational analysis.** My research focuses on how neuromodulatory signals shape dendritic computation in the apical dendrites of layer 5 pyramidal neurons. I use SCAPE, two-photon, and random-access microscopy to record dendritic calcium and neuromodulatory dynamics in awake mice, and I develop custom (Python, MATLAB) pipelines to identify and classify dendritic events.

Education

Ph.D. in Neurobiology — **Boston University**
Expected May 2027

Neurophotonics / Biomedical Engineering focus

M.Sc. in Cell and Molecular Biology — **Boston University**
2022 – 2024

B.Sc. in Cell and Neurobiology — **J. W. Goethe University Frankfurt**
2018 – 2021

Thesis: Neurovascular Interface in Aging Zebrafish

Work Experience

Boston University — September 2022 – Present
Ph.D. Candidate, Neurovascular Imaging Laboratory (Devor Lab) — Boston, MA

- *Dissertation: Behavioral-State-Dependent Calcium Dynamics in the Apical Dendrites of L5 Pyramidal Cells.*
- Assembled a **SCAPE light-sheet microscope** for volumetric Ca^{2+} imaging of L5 apical dendrites in awake mice.
- Applied **random-access two-photon microscopy (Femto3D Atlas, Femtonics)** to image volumetrically from soma to apical dendrite, distinguishing local dendritic events from somatic spiking.
- Developed a **custom Python toolbox** for dendritic event detection, showing how Ca^{2+}-spike probability scales with acetylcholine.

Brigham and Women's Hospital, Harvard Medical School — July 2023 – September 2023
Summer Research Trainee, Neurology Department (Smirnakis/ Venkatesh Lab) — Boston, MA

- Designed and set up an in vivo **two-photon imaging** approach in awake mice to study tumor-associated epilepsy.
- Developed **Python** and **ImageJ** pipelines to preprocess and analyze imaging data.

Brigham and Women's Hospital, Harvard Medical School September 2021 – February 2023

Research Intern, Neurology Department (Smirnakis Lab) Boston, MA

- Developed a **visual bistable perception paradigm** for simultaneous behavioral and two-photon imaging in mice, improving experimental control and reproducibility.
- Built and maintained **semi-automated analysis software Dolia** to accelerate OKN time-series processing.

Institute for Cell Biology and Neuroscience, Goethe University Frankfurt March 2020 – May 2021

Research Intern, Molecular & Cellular Neurobiology (Acker-Palmer Lab) Frankfurt, Germany

- Conducted **single-cell microinjections** and advanced microscopy (confocal, two-photon, light-sheet) to study axonal regeneration in zebrafish spinal-cord injury models.
- Designed and supervised a **Master-level teaching module** introducing neural injury models and microscopy workflows.

Max Planck Institute for Brain Research July 2019 – March 2020

Research Intern, Electron Microscopy Unit Frankfurt, Germany

- Reconstructed 3-D neuronal morphology from serial electron-microscopy data using **KNOSSOS** to map microcircuit connectivity.
- Collaborated with data-analysis engineers to optimize segmentation and tracing speed through software-parameter tuning.

Publications

Bogatova, D., Smirnakis, S., Palagina, G.	*Tug-of-peace: Visual Rivalry and Atypical Visual Motion Processing in MECP2 duplication syndrome of autism.* ENeuro (2023). https://doi.org/10.1523/ENEURO.0102-23.2023
Herrema, K. E., Kharitonova, E. K., **Bogatova, D.** *et al.*	*A neurorecording toolkit for longitudinal assessments of transplanted human cortical organoids in vivo.* BioRxiv (2025). https://doi.org/10.64898/2025.12.20.695690
Hu, G., Deng, Q., Qi, T., Chen, Z., Rauscher, B. C., Chai, N., **Bogatova, D.** *et al.*	*Bio-CM2: Distributed computational optics for cortex-wide cellular imaging.* BioRxiv (2026). https://doi.org/10.64898/2026.07.27.740823
Fomin-Thunemann, N., Martin, E., Thunemann, M., Rauscher, B., **Bogatova, D.** *et al.*	*Effect of focused ultrasound blood-brain barrier disruption on neurovascular coupling.* (in preparation)

Bogatova, D. *et al.*	*Behavioral-State-Dependent Calcium Dynamics in the Apical Dendrites of the L5 Pyramidal Cells.* (in preparation)

Poster Presentations

2025 *Volumetric Imaging of Resting State Calcium Dynamics in Apical Dendrites*

- Society for Neuroscience Annual Meeting, San Diego, CA
- New England Society for Microscopy Fall Symposium
- Frontiers in Neurophotonics Summer School

Laboratory Skills

Microscopy	**SCAPE**, **Two-Photon**, Confocal, Light Sheet, Mesoscope, Widefield, Electron Microscopy, **Random-Access Microscopy (Femto3D Atlas, Femtonics)**
Molecular & Histological	Immunohistochemistry, In-Situ Hybridization, qPCR, Tissue Clearing, Cryo-/Vibratome Sectioning, H&E Staining
In Vivo Techniques	**Mouse Surgeries**, Viral Vector & **Retro-Orbital (PHP.eB)** Injections, Cranial Window Preparation, **Awake Head-Fixed Imaging**, Behavioral Monitoring (pupillometry, whisking, accelerometry), Pup Injections

Technical Skills

Programming & Analysis	**Python** (NumPy, SciPy, scikit-image, pandas, matplotlib, zarr), **Agentic AI** (Claude, Gemini, ChatGPT), MATLAB, R, Git, Jupyter, UNIX Shell
Image & Data Processing	**Fiji/ImageJ**, **Napari**, Imaris, CaImAn, DeepLabCut, Suite2P, KNOSSOS
Visualization & Design	Adobe Illustrator, Photoshop, Premiere Pro, Procreate, PowerPoint
Web & Cloud	HTML/CSS, Bash, GitHub, Overleaf

Honors and Awards

2026	1st Place, Research on Tap, Boston University Photonics Center
2025	Neurophotonics Research Trainee, Boston University
2022	Dean's Fellowship, Boston University
2022	Chancellor's Fellowship, UC Santa Barbara
2021	Certificate of Achievement in Immuno-Oncology, Harvard Medical School

Languages

Ukrainian	Native
English	Advanced (TOEFL 113/120)
German	Advanced (TestDaF 5/5)
russian	Native

Leadership and Activities

Mriya Inc. **May 2022 – Present**
President & Co-Founder Boston, MA

Founded and lead **Mriya Inc.**, a 501(c)(3) nonprofit organization supporting humanitarian and medical aid for Ukraine (EIN 88–2899667).

Ukraine Support

- Raised over **$300,000** through fundraising events, partnerships, and online campaigns.
- Delivered critical aid such as medical supplies, drones, and evacuation vehicles to frontline brigades.
- Oversee strategic planning, donor relations, and public communications across international teams.
- Coordinate collaborations with other NGOs to ensure transparent logistics and reporting.

Pediatric Oncology

- Medical-document translation (Ukrainian / English / German);
- Outreach and patient referrals to U.S. and European (German) clinics;
- Dedicated fundraising for treatment and travel costs;
- Family emotional support and end-of-life care.

"Krab" Volunteering Organization March 2014 – Present
Volunteer Coordinator Kyiv, Ukraine

- Since 2022, work with Kyiv hospitals to identify **pediatric oncology patients** and initiate cross-border transfers for treatment abroad.
- Lead local fundraising and hospital-outreach initiatives since 2014, sustaining continuous supplies and volunteer engagement.